School of Computer Science and Engineering

Department of Computer Engineering and Technology

T. Y. B. Tech CSE **AY 2026-27** **Semester 5**

Computer Networks Lab (CSE20110)

CN Lab Assignment: 6

Title: Write a program UDP Socket Programming: (Language: C/C++/Python/Java).

Aim: Write a program for wired network using UDP socket to perform any one of the following operations

a. String Conversion from UpperCase to LowerCase (50% students)
b. Reverse the String (Remaining 50% students)

Objectives:

1. To understand concept of socket programming using UDP

Theory: (Handwritten & Scanned)

- Students should write following theory

1. UDP Header.
2. UDP Socket Primitives at Server and Client

Pseudocode:

- Server program-
  1. START
  2. Create UDP socket
  3. Bind socket to IP address and port number
  4. Wait to receive message from client
  5. Receive message and client address
  6. Display received message
  7. Send reply to client address
  8. Close server socket
  9. STOP
- Client Program -
  1. START
  2. Create UDP socket
  3. Input message from user
  4. Send message to server IP address and port number
  5. Receive reply from server
  6. Display server reply
  7. Close client socket
  8. STOP

Program Code (for programming assignment) / CLI code (for Packet Tracer Assignments):

- Students should write/paste code here

Output Screenshots:

- Students should Draw/Paste final output screenshots

Students Observation:

- Students should write their observations about lab assignment

FAQ (Handwritten & Scanned)

- Students should write answer for following questions

1) What is UDP and why is it used in socket programming?
2) What are the important functions used in UDP socket programming ?
3) What is the difference between TCP and UDP socket programming with respect to connection establishment, reliability, flow control, congestion control, speed? Give 2 real world examples.?
4) What happens if a UDP packet is lost during transmission?
5) State five applications of UDP and What is Ephemeral Ports?